

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

T. Lakshay (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

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20 Artificial Intelligence.

1. Smart Home Automation System using Intelligent Agents:

An Intelligent Agent is a system that perceives its environment using sensors and performs action using actuators to achieve specific goals.

Smart Home Automation. - Sensors

- * Motion
- * Temperature
- * Light Intensity
- * Time of day.
- * Door / window status.

The system controls are

- * Lights.
- * Air Conditions
- * Heater
- * Security alarms
- * Door locks.

1. Simple Reflex Agent.

A Simple Reflex Agent acts only on the current perception using predefined IF - THEN rules.

It does not remember previous events.

Ex: $\Rightarrow$ Turns on the light when there is motion.

$\Rightarrow$ If temperature is greater than 30°C turn on AC.

2. Model-Based Agent

A Model Based Agent maintains an internal model of the environment and remembers previous states. It are occupied continuously updates its knowledge based on sensor inputs.

$\Rightarrow$ Maintains an internal memory

$\Rightarrow$ Remembers previous states.

$\Rightarrow$ Handles partially observable environments.

3. Goal Based Agent.

A Goal Based Agent makes decision by considering a specific goal that need to be achieved. Instead of home, the goals may include maintaining a comfortable room temperature reducing electricity consumption.

$\Rightarrow$ Improve home security.

2. Robot in an Unknown maze using Uniformed Search strategies.

* A robot placed inside a unknown maze must search for the exit without having any prior knowledge of the maze structure.

* This problem can be solved using uniform search algorithm, because the robot has no information about the location of the goal.

* Two commonly used uniform search algorithm are

i) Uniform Cost search (UCS)

ii) Iterative Deepening Search.

~~1. UCS~~

Example:

Path A = Cost 5

Path B = Cost 8

Robot selects Path A because it has the lowest cost.

1. Uniform Cost Search (UCS)

⇒ UCS is a uniformed Search algorithm that always expands the nodes with the lowest cumulative path cost.

⇒ Instead of selecting the shortest path based on the no. of steps, use chooses the path with minimum total cost.

⇒ Therefore it guarantees finding the optimal solution when all cost are positive.

Iterative Deepening Search.

* Iterative Deepening Search combines the advantages of Depth first Search and Breadth first Search.

* It repeatedly performs Depth-Limited Search by increasing the search depth one level at a time until the goal is found.

* IDS is very useful when the depth of the goal is unknown because it requires less memory.

Ex:

Depth 0 → Search Root

Depth 1 → Search Children

Depth 2 → Search Grandchildren.

Continues until Exit is found.

Advantages & Disadvantages.

UCS

- Finds the least cost-path
- Complete algorithm
- Optimal solution
- This are the advantages of uniform cost search.

IDS

- Uses very little memory
- Complete algorithm
- Suitable for when the goal ~~at~~ depth is unknown.

Time & Space Complexity

Algorithm	Time Complexity	Space complexity
Uniform cost search	High (Depend on path cost)	High
Iterative Deepening search.	$O(b^d)$	$O(b \times d)$

b - branch factor
d - depth.

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand